

SIDHARTH PADMANABHAN

+41 0779699586 | sidharthpadman9@gmail.com | linkedin.com/in/sid-wc121 | sid-wc121.github.io

EDUCATION

Università della Svizzera italiana (USI)

MSC IN INFORMATICS

Lugano, Switzerland

2025 – Present

- Relevant Courses: Distributed Systems, Graph Deep Learning, Data Analytics, Robotics, Advanced ML

Amrita Vishwa Vidyapeetham

B.TECH. ELECTRONICS AND COMPUTER ENGINEERING

Amritapuri, Kerala, India

2021 – 2025

- Relevant Courses: Data structures & algorithms, Calculus, Data Science, Deep Learning, Computer Networks

PROFESSIONAL EXPERIENCE

Singapore University of Technology and Design (SUTD)

RESEARCH INTERN - ROBOTICS

Singapore

Feb 2024– Apr 2024 | Onsite

- Tested and helped re-implement Smorphi educational robot curriculum designed by the SUTD research team
- Assisted in inventory management and quality control at **Wefaa Robotics** (15 robots), a startup spin-off from SUTD
- Led the development of a Smorphi-based cleaning robot, focusing on LiDAR mapping, navigation, and coverage optimization
- Worked on navigation backbone that allowed seamless integration of sensor controls, resulting in ~30% faster performance

HuT Labs

RESEARCH INTERN - MACHINE LEARNING

Kerala, India

Aug 2023 – Feb 2024 | Onsite

- Re-implemented ML components from multiple research papers including using SVM to fuse MobileNet SSD and MobileNetv2 outputs for staircase detection in "sTetro" robot project
- Collected, processed, and labeled data (9500+ samples) to train and evaluate models for staircase detection
- Developed a custom deep CNN model with 23% improvement in staircase detection over other benchmarked models

Team bi0s

FULL-STACK DEVELOPER

Kerala, India

Oct 2022 – Jan 2023 | Hybrid

- Collaborated with cross-functional teams to design and develop a CMS (Content Management System) website
- Implemented responsive web pages and multi-tier user login using Vite-React.js, Redux, and Tailwind CSS
- Gained agile development experience through sprint planning and stand-up meetings

PUBLICATIONS

(Google Scholar)

- Manikandan Arunachalam, **Sidharth P**, Sooraj Nair, Nikhil Rajiv, Rakesh Suresh Babu. (2025) "Blockchain Approach for Real-Time Authentication in 6G Machine-to-Machine Communication". In: **IGI Global**. Book: **6G Networks and AI-Driven Cybersecurity**, Shaikh, Riaz Ahmed. (Link)
- **Sidharth P**, Dr. Rajesh Kannan Megalingam. (2024) "Custom Deep CNN-Based Staircase Classification for Smart Environments". In: **ICONAT 2024**. (Link)
- Sreedutt Ram J, **Sidharth P**, Sukrith Sunil, Poorna S S, Anuraj K. (2024) "Comparative Analysis of DL Models Using Cough Audio Data for Early Detection of COVID-19". In: **ICCIDA 2024**. (Link)
- Manikandan Arunachalam, **Sidharth P**, Sooraj Nair, Nikhil Rajiv, Rakesh Suresh Babu, Sanjay T. (2024) "Blockchain-based IoT Computing Services in Medical Healthcare Solutions - a thorough review". In: **IGI Global**. Book: **Blockchain-Based Solutions for Accessibility in Smart Cities**, Kumar Abhishek, Chinmay Chakraborty. (Link)

PROJECTS

(GitHub)

Vision-Based Gripper Aperture Estimation | Pytorch, Lua, ROS2, CoppeliaSim

Apr 2026 – June 2026

- Automated data collection with gripper prismatic joint positions using CoppeliaSim's ZMQ API, bypassing ROS joint-state.
- Built concurrent ROS 2 executor for simultaneous autonomous navigation and gripper sweep, for background-diverse data.
- Trained a lightweight regression CNN (0.015 MAE) to estimate 0–87 mm fingertip aperture from a single RGB frame.
- Deployed live inference node with predicted vs ground-truth overlay, rolling error chart.

Paper Replication: "Understanding Inter-Concept Relationships in Concept-Based Models"

May 2026

- Reproduced BAIC (ICML 2024) experiments across Colored MNIST, dSprites, CUB, and CheXpert datasets.
- Built automation pipelines for Concept2Vec, TCAV, CEM, metric evaluation, and reproducible experiment tracking.

- Fixed theoretical and code base inconsistencies, training pipeline issues, and missing components for reliable execution.
- Proposed a tensorial semantic coupling formulation, generalizing concept intervention as context-aware message passing.

Visual Notebook V2 [🔗] | *Polars, DuckDB, React, TypeScript*

Apr 2026 – May 2026

- Built a no-code visual analytics platform using React, DuckDB, and Polars for interactive exploration of tabular datasets.
- Designed a git tree architecture supporting state tracking, branching workflows, and reproducible data transformations.
- Implemented a modular visualization engine generating ECharts specifications from backend APIs.
- Evolved the platform from an earlier notebook style prototype (**Visual Notebook V1**), redesigning the frontend.

Spatiotemporal GNN for Probabilistic Postprocessing of Wind Speed Forecasts [🔗] | *PyTorch*

Nov 2025 – Dec 2025

- Re-designed a Temporal Graph MLP-Mixer for weather forecasting to predict 97hr lead times at each stations in parallel.
- Attained 0.9017 MAE further reduced to 0.88 MAE on optimization, 0.6243 CRPS with <0.6% variance across 5 seeds.
- Trained under 35 min on RTX3070, outperforming heavy graph/attention based models requiring >1hr on A100 GPU.

Datacenter Simulator [🔗] | *Python*

Nov 2025 – Dec 2025

- Built a discrete event datacenter simulation (priority queues) with deterministic, reproducible testing.
- Modeled realistic network physics with leaf-spine topology, congestion-aware routing, partitions, and latency calculation.
- Implemented hardware failure injection with resource fluctuation and Chaos Monkey-style node shutdowns
- Simulated distributed time challenges including clock drift and synchronous/async communication with timeout handling.
- Extended into a **protocol testing suite** used to validate consensus algorithms (Paxos, Raft) and replication protocols under controlled fault injection.

Medical Image Anomaly Detection and Classification | *PyTorch*

Mar 2025 – June 2025

- Curated a unified dataset combining Brain MRI, Chest X-ray, and Liver CT images for anomaly detection and classification.
- Used ResNet-50 (Blocks 2 & 3) for feature extraction in an autoencoder-based adversarial network setup.
- Achieved clear anomaly separation on brain and lung data; liver CT showed moderate separation.
- Reached 87% classification accuracy across brain, lung, and liver disease classes.

Biomarker Identification for Diabetic Nephropathy | *sklearn, TensorFlow*

Oct 2024 – Nov 2024

- Trained autoencoders with 250 perceptrons in the intermediary layer and 21 in the bottleneck layer.
- Generated heatmaps for raw data, intermediate, and bottleneck layers to analyze encoding.
- Selected 30 unique genes using a 0.7 threshold on normalized cumulative counts.
- Applied RFE with SVM as the base classifier to reduce to 19 genes, achieving the best classification accuracy.

HFCC Feature Extraction Package | *Python*

Dec 2023 – Jan 2024

- Developed an HFCC (Human Factor Cepstral Coefficients) feature extraction package supporting the ICCIDA paper "*Comparative Analysis of DL Models Using Cough Audio Data for Early Detection of COVID-19*".
- Designed the package to be fully customizable, allowing users to adjust each parameter according to specific requirements.

LANGUAGES

Malayalam Mother tongue; **English** Fluent; **Hindi** Basic; **Tamil** Basic

TECHNICAL SKILLS

Programming: Python(proficient), R(proficient), C(intermediate), JavaScript(intermediate), Matlab(intermediate)

Frameworks: Tensorflow(proficient), PyTorch(proficient), React.js(advanced), Next.js(advanced), Node.js(intermediate)

POSITIONS OF RESPONSIBILITY

IEEE RAS Chair at Student Branch, Amrita Vishwa Vidyapeetham

Apr 2024 – Jan 2025

- Sharing information about funded projects in collaboration with IEEE, and providing learning resources
- Organizing robotics-related talks and workshops, fostering student engagement in robotics and ML

Lead Colorist, Media Amritapuri (now B Rollers)

Oct 2023 – Mar 2025

- Responsible for color correction and enhancement of media content, ensuring high-quality visual output. Collaborated with creative teams to achieve visual excellence in images.

Executive committee member, Vidyut (campus multi-fest)

Dec 2023 – Apr 2025

- Responsible for photography, videography quality maintenance, managing junior colorist team and recruiting new members.

EXTRA-CURRICULAR ACTIVITIES

SSR Project [🔗] - Hosted a 3 day Technical workshop to impart knowledge and skill to college students on Cyber Security, Microcontrollers (Arduino), C programming, Version Control Systems (Git/Github) and Conducted a coding contest for them as part of Student's Social Responsibility (SSR) initiative.

Community Services - AmritaSREE Kit Distribution Seva [🔗] - Engaged in community service by participating in the AmritaSREE Self Help Groups organised by Mata Amritanandamayi Math on 24th December 2022 at Cherthala